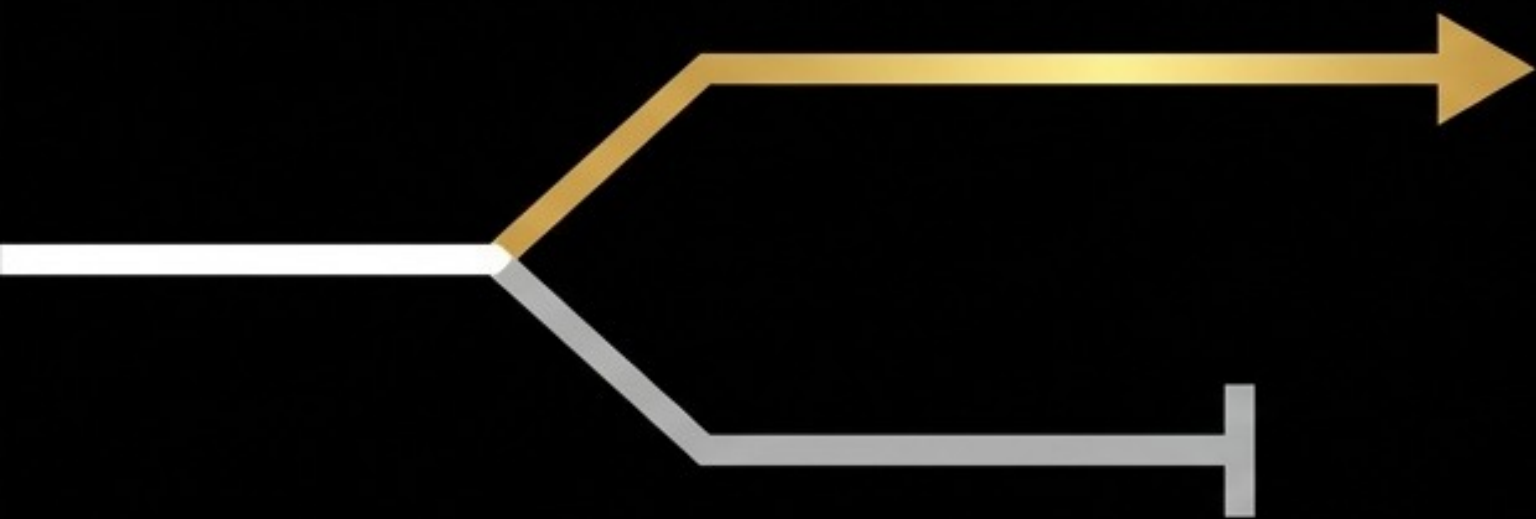




FORKS, FINALITY, AND REORGANIZATIONS

ARCHITECTURE AND STATE (02.4)

In a network without a central clock or master server, chain fractures are inevitable. This module dissects why forks occur, how the network auto-recovers from divergence, and when a transaction achieves permanent finality.

THE TWO FACES OF “FORK”

1

CONSENSUS RULE UPGRADES

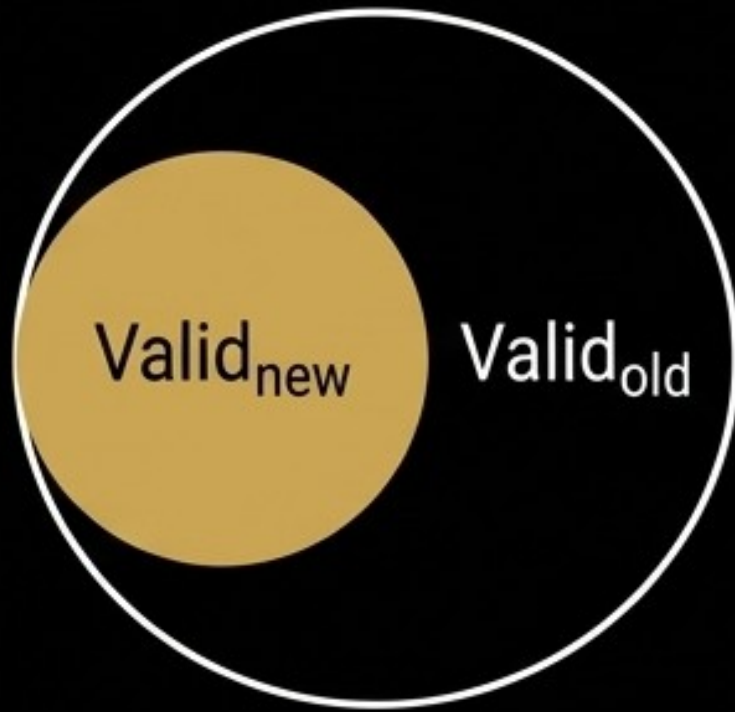
- Intentional or debated modifications to client software validation rules.
- Driven by developer consensus and human coordination.
- Categorized **strictly into Soft Forks and Hard Forks**.

2

TRANSIENT STATE DIVERGENCES

- Accidental chain splits caused by **data propagation delays** in the peer-to-peer network.
- Driven by physical limits (latency and the speed of light).
- Results in **Block Reorganizations (Reorgs)** and orphaned blocks.

CONSENSUS RULE UPGRADES

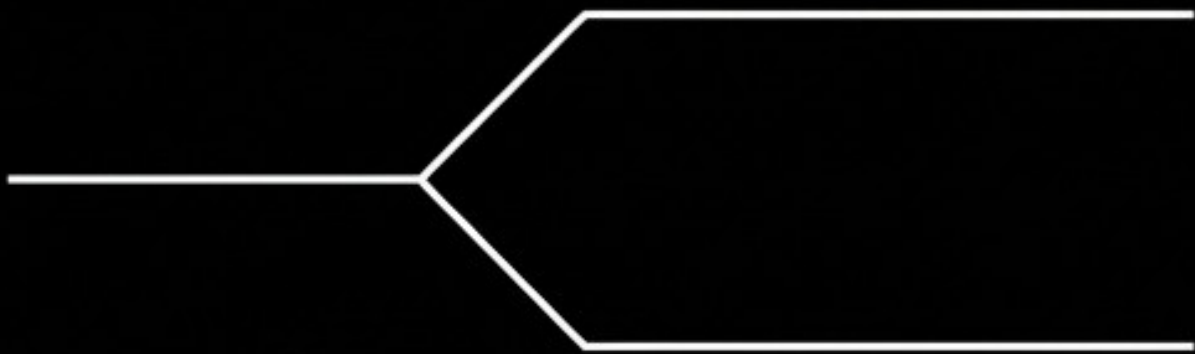


1. SOFT FORKS (BACKWARD COMPATIBLE)

$$\text{Valid}_{\text{new}} \subset \text{Valid}_{\text{old}}$$

Rules are restricted to a stricter subset.
Old nodes view new blocks as valid.

BIP-66 (2015), SegWit (2017), Taproot (2021).



2. HARD FORKS (NON-BACKWARD COMPATIBLE)

$$\text{Valid}_{\text{new}} \not\subset \text{Valid}_{\text{old}}$$

Rules are expanded or fundamentally altered.
Old nodes reject new blocks entirely.

2013 Bitcoin DB Bug, The DAO (2016), The Merge (2022).

THE PHYSICS OF LATENCY



SIMULTANEOUS DISCOVERY

Miner A (Iceland) and Miner B (Singapore) discover a valid block at height N on the exact same millisecond.

PROPAGATION DELAY

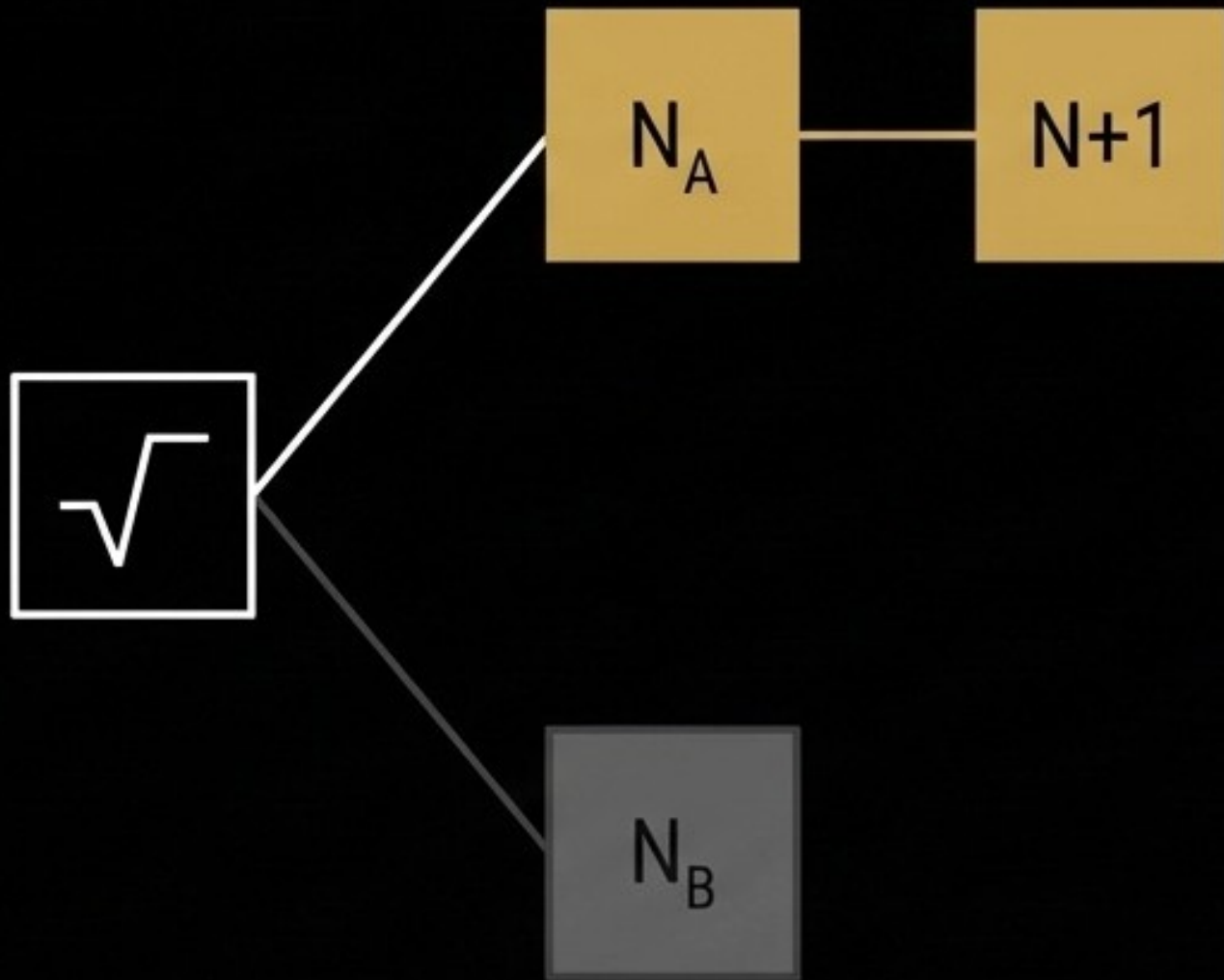
It takes 100–200 milliseconds for light signals to traverse intercontinental submarine cables.

European nodes receive N_A first.
Asian nodes receive N_B first.

RESULT

The global network briefly fractures into two equally valid historical realities.

ANATOMY OF A REORG



A Poisson Process breaks the tie. Mining is purely random, making simultaneous discovery of block $N+1$ nearly impossible.

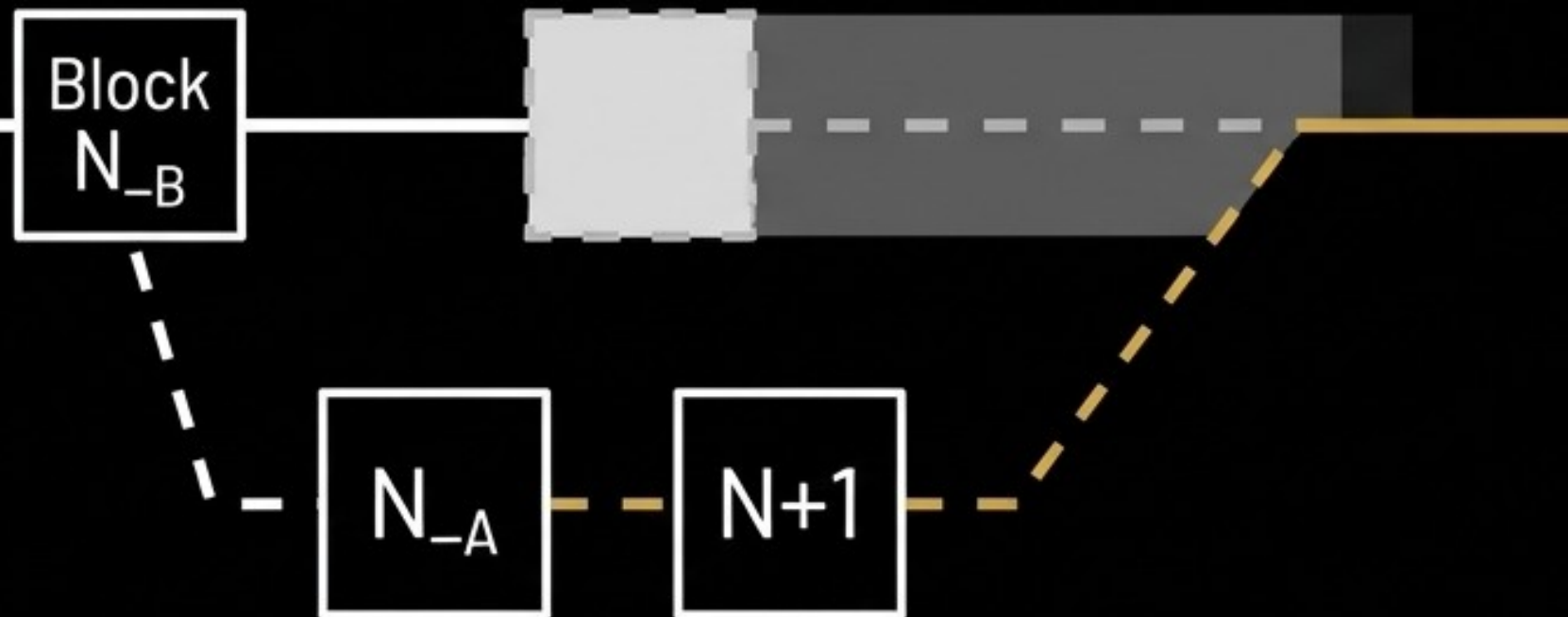
1 Tie-Breaker: Miner C discovers Block $N+1$ built atop N_A (Iceland branch) and broadcasts it globally.

2 Evaluation: Asian nodes receive the new chain. The $[N_A \rightarrow N+1]$ branch has two blocks of Proof of Work. Their local $[N_B]$ branch has one.

3 Execution (Reorg): Nodes strictly obey consensus. They **rollback** state mutations from N_B and apply states from N_A and $N+1$.

4 Orphaning: Block N_B becomes an Orphaned Block. Its unique transactions return to the mempool.

THE REORG DOUBLE-SPEND



Mallory's 1-Confirmation Attack

Step 1: Mallory deposits 1,000 coins into an exchange within public Block N_B .

Step 2: The exchange carelessly assumes 1 block confirmation is final and allows Mallory to withdraw fiat cash.

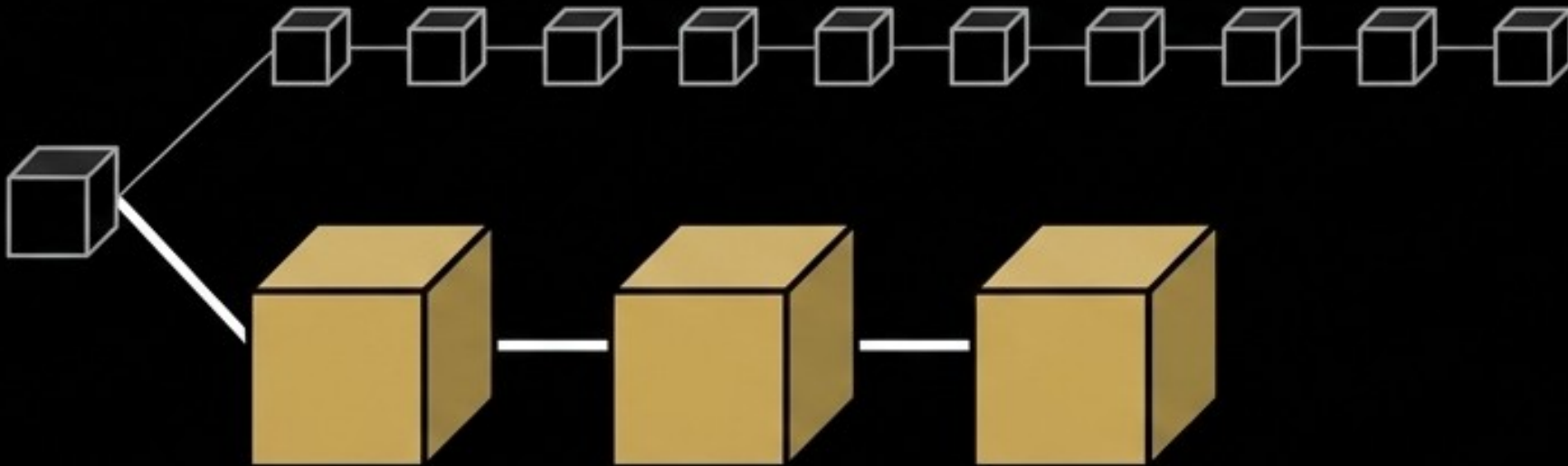
Step 3: Secretly, Mallory mines a private chain $[N_A \rightarrow N+1]$ routing those 1,000 coins to his own wallet.

Step 4: Mallory broadcasts his heavier private chain. The network executes a Reorg.

Step 5: Block N_B is orphaned. The exchange deposit record is erased from official history. The exchange loses 1,000 coins.

THE FORK-CHOICE RULE

$$\text{Canonical Chain} = \operatorname{argmax}(\text{chain}) \sum \text{Difficulty}(B_i)$$



THE HEAVIEST-CHAIN RULE

Popularly called the “longest-chain rule,” but nodes actually calculate total accumulated Proof of Work difficulty.

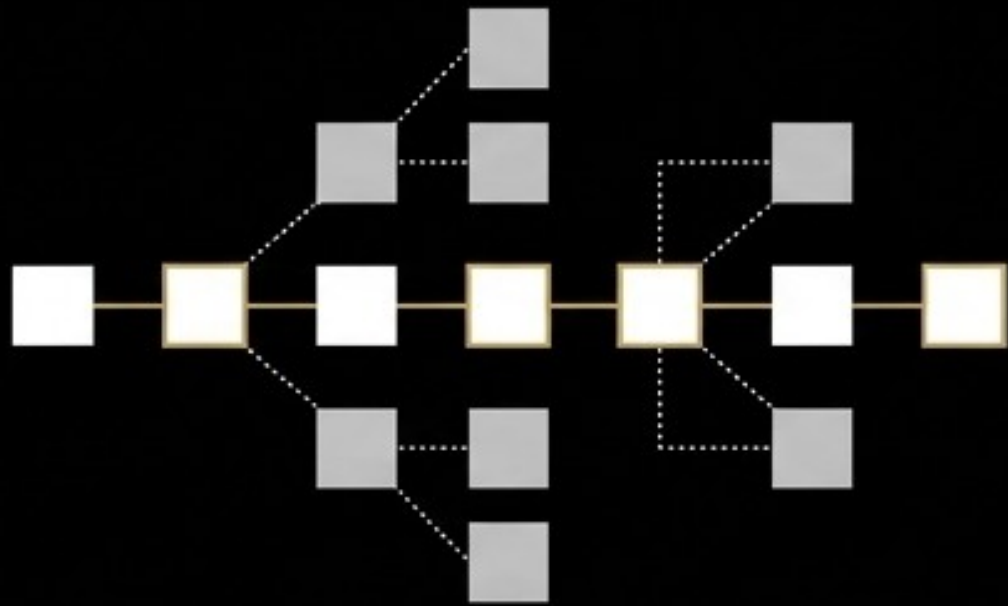
PREVENTING EXPLOITS

If nodes measured raw block height, an attacker could instantly forge millions of fake blocks at zero difficulty.

THERMODYNAMIC ANCHOR

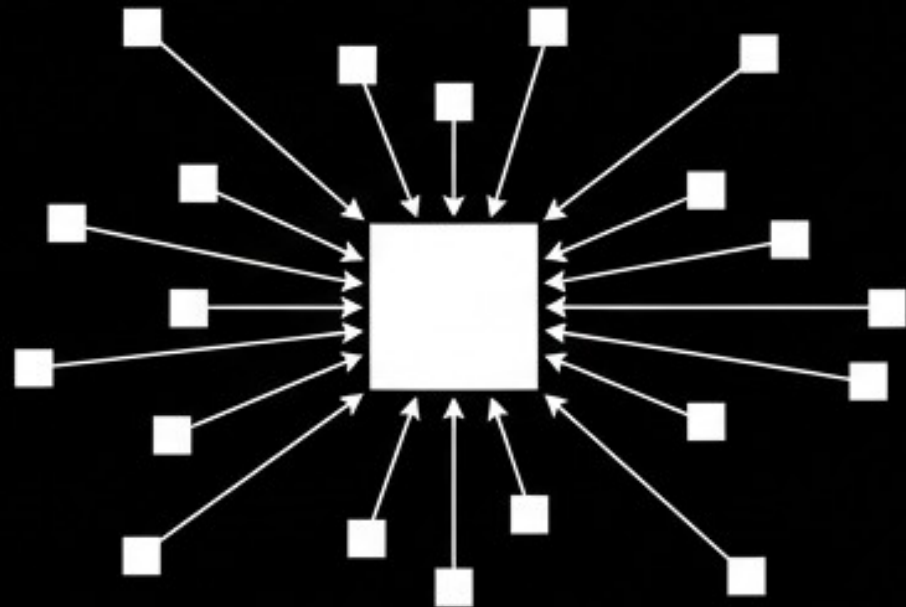
By summing difficulty, nodes guarantee the canonical chain represents the greatest sacrifice of real-world thermodynamic energy.

FORK-CHOICE EVOLUTION



GHOST PROTOCOL (Ethereum Era PoW)

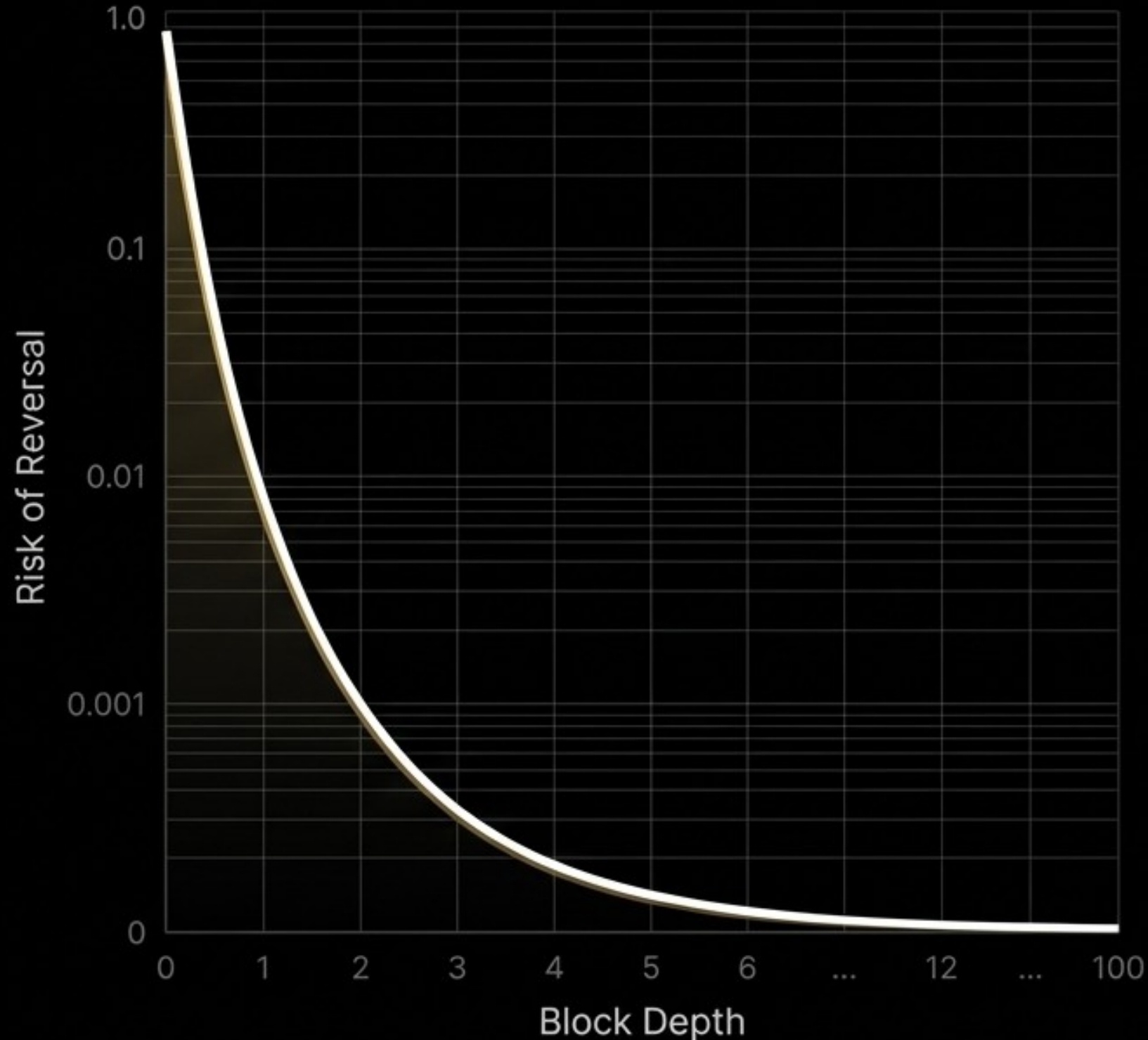
Ethereum's fast 12-second block times caused orphaned block rates to spike to 10%. Greedy Heaviest Observed Sub-Tree integrated orphans as ommer/uncle blocks. They contributed security weight to the main chain and yielded partial miner subsidies (75%–87.5%).



LMD-GHOST (Ethereum Era PoS / Gasper)

Consensus shifted from thermodynamic hashing power to cryptographic stake attestations. Latest Message Driven calculates only the most recent attestation from active validators. Nodes choose the child block accumulating the greatest total volume of deposited validator stake.

PROBABILISTIC FINALITY (PoW)



In Nakamoto Consensus, finality is never mathematically absolute; it is **probabilistic**.

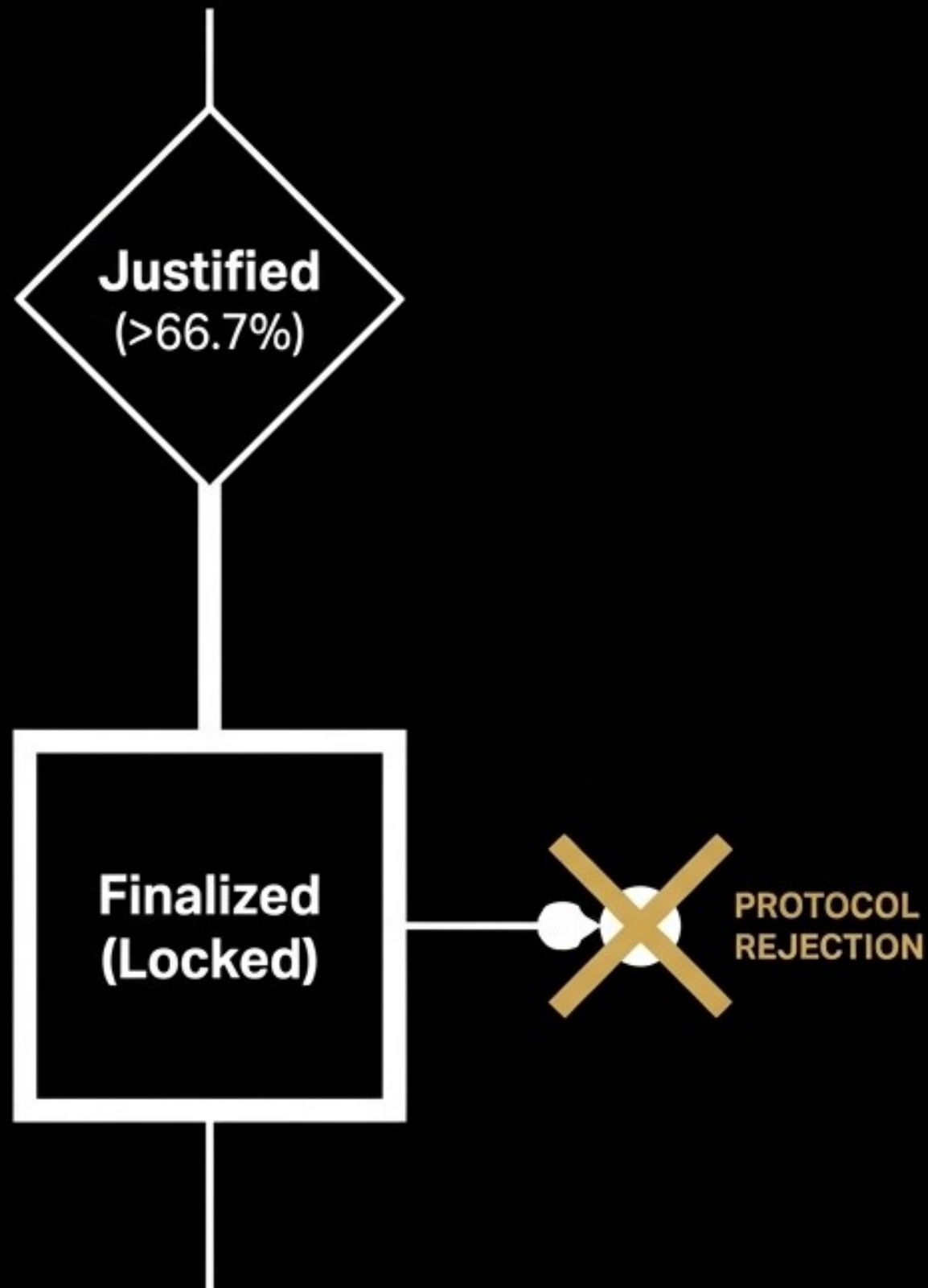
$$P \approx \sum \frac{\lambda^k e^{-\lambda}}{k!} \left(\frac{q}{p} \right)^{\max(z-k, 0)}$$

An attacker with minority hash power ($q < 0.5$) faces an exponentially shrinking probability of catching up to the honest chain.

THE 6-CONFIRMATION STANDARD

- After 6 blocks (~1 hour in Bitcoin), the probability of a 10% hash-rate attacker reversing a transaction drops below 0.1%.
- Against a 30% attacker, the risk shrinks below 1.3%.
- The theoretical risk of a Reorg remains eternally present under a 51% attack.

DETERMINISTIC FINALITY (PoS)



THE EPOCH PIPELINE (Casper FFG)

1. **Justification:** Achieved when $> 66.7\%$ of total validator stake cryptographically signs an epoch checkpoint.
2. **Finalization:** When the subsequent checkpoint is justified, the preceding checkpoint officially achieves Finalized status.

THE ECONOMIC ANTI-REORG GUARANTEE

Once finalized, full nodes will never reorganize that block under any network condition. To mathematically force two conflicting finalized checkpoints, $> 33.3\%$ of global validators must cryptographically vote twice.

The protocol will automatically detect the fraud, instantly seizing and burning billions of dollars in attacker stake.